CSL7110: Machine Learning with Big Data

Assignment 2

Min-Hashing and Locality Sensitive Hashing

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GitHub Repository:
https://github.com/Jyoti-Dwivedi-010/ML_with_Big_Data_Assignment2

1 Question 1: K-Grams (20 Points)

1.1 Problem Statement

Construct several types of k-grams for documents D1–D4:

- Character 2-grams
- Character 3-grams
- Word 2-grams

Each k-gram is stored only once (duplicates are ignored). Then compute the Jaccard similarity for all $\binom{4}{2} = 6$ document pairs for each gram type, giving $3 \times 6 = 18$ values.

1.2 Methodology

- **Character k-grams:** Sliding window of size k over the raw text. Space counts as a character.
- **Word k-grams:** Sliding window of size k over whitespace-tokenized words.
- **Jaccard similarity:** $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$

1.3 1A: K-Gram Construction

The following table shows the number of unique k-grams for each document and gram type:

```
--- Character 2-grams ---
  Document  Unique k-grams   Sample (first 5)
  ------------------------------------------------------------
  D1                   263   ' a', ' b', ' c', ' d', ' e'
  D2                   262   ' a', ' b', ' c', ' d', ' e'
  D3                   269   ' a', ' b', ' c', ' d', ' e'
  D4                   255   ' a', ' b', ' c', ' d', ' e'
```

```
--- Character 3-grams ---
  Document  Unique k-grams   Sample (first 5)
  ------------------------------------------------------------
  D1                   765   ' a ', ' ab', ' ac', ' ad', ' ai'
  D2                   762   ' a ', ' ab', ' ac', ' ad', ' ai'
  D3                   828   ' a ', ' ab', ' ac', ' ad', ' ag'
  D4                   698   ' a ', ' ab', ' ac', ' ad', ' af'
```

```
--- Word 2-grams ---
  Document  Unique k-grams   Sample (first 5)
  ------------------------------------------------------------
  D1                   279   ('a', 'difficult'), ('a', 'hockey'), ('a',
  D2                   278   ('a', 'difficult'), ('a', 'hockey'), ('a',
  D3                   337   ('a', 'differentiating'), ('a', 'difficult'
  D4                   232   ('a', 'candidate'), ('a', 'prosperous'), ('

--- Uniqueness Verification (D1, Character 2-grams) ---
  Total 2-grams (with duplicates): 1748
  Unique 2-grams (after dedup):    263
  Duplicates removed:              1485
  -> Confirms that duplicate k-grams are correctly ignored.
```

Figure 1: K-gram construction results showing unique k-gram counts for all documents.

1.4 1B: Jaccard Similarity Results (18 Numbers)

Table 1: Jaccard Similarity for All Document Pairs and Gram Types

Pair	Char 2-gram	Char 3-gram	Word 2-gram
D1–D2	0.9811	0.9780	0.9408
D1–D3	0.8157	0.5804	0.1823
D1–D4	0.6444	0.3051	0.0302
D2–D3	0.8000	0.5680	0.1737
D2–D4	0.6413	0.3059	0.0303
D3–D4	0.6530	0.3121	0.0161

```
============================================================
  1B: JACCARD SIMILARITY RESULTS
  3 gram types × 6 pairs = 18 numbers
============================================================

=== Character 2-grams ===
  Pair         Jaccard Similarity
  ----------------------------------
  D1-D2                  0.9811
  D1-D3                  0.8157
  D1-D4                  0.6444
  D2-D3                  0.8000
  D2-D4                  0.6413
  D3-D4                  0.6530

=== Character 3-grams ===
  Pair         Jaccard Similarity
  ----------------------------------
  D1-D2                  0.9780
  D1-D3                  0.5804
  D1-D4                  0.3051
  D2-D3                  0.5680
  D2-D4                  0.3059
  D3-D4                  0.3121
```

```
============================================================
  SUMMARY TABLE: All 18 Jaccard Similarity Values
============================================================

  Pair        Char 2-gram  Char 3-gram  Word 2-gram
  ---------------------------------------------------
  D1-D2          0.9811       0.9780       0.9408
  D1-D3          0.8157       0.5804       0.1823
  D1-D4          0.6444       0.3051       0.0302
  D2-D3          0.8000       0.5680       0.1737
  D2-D4          0.6413       0.3059       0.0303
  D3-D4          0.6530       0.3121       0.0161
```

Figure 2: Jaccard similarity results for all 18 document pair/gram type combinations.

1.5 Observations

1. **D1 and D2** have the highest similarity across all gram types (∼0.93–0.98), as they are near-duplicate documents with minor textual changes.

2. **D1–D3 and D2–D3** show moderate similarity (∼0.36–0.58), covering similar topics (Apple/Cook) with different wording.

3. **D4** has low similarity with all other documents (∼0.10–0.31), being about a completely different topic (Trump/Davos).
4. **Character 2-grams** produce the highest similarity values because short character sequences are commonly shared across English text.
5. **Word 2-grams** produce the lowest similarity because exact word pair matches are harder to achieve.
6. The ordering is consistent: J(D1,D2) $>$ J(D1,D3) $\approx$ J(D2,D3) $>$ J(D3,D4) $>$ J(D1,D4) $\approx$ J(D2,D4).

2 Question 2: MinHashing (20 Points)

2.1 Problem Statement

2A: Use character 3-grams to build MinHash signatures for D1 and D2 with $t = 20, 60, 150, 300, 600$ hash functions. Report the approximate Jaccard similarity for each t.

2B: Determine a good value of t by analyzing accuracy versus computation time.

2.2 Methodology

We use a universal hash family H where each $h \in H$ maps:

$$h(x) = (ax + b) \mod m, \quad m = 100{,}003 \text{ (prime, } > 10{,}000)$$

The MinHash estimator is:

$$JS_t(a, b) = \frac{1}{t} \sum_{i=1}^{t} \mathbf{1}[a_i = b_i]$$

2.3 2A: MinHash Similarity Estimates (5 Numbers)

Table 2: MinHash Estimated Jaccard Similarity for D1 vs. D2 (Character 3-grams)

t	**Approx. Jaccard**	**Absolute Error**
20	1.0000	0.0220
60	0.9667	0.0113
150	0.9933	0.0154
300	0.9767	0.0013
600	0.9800	0.0020

```
============================================================
  QUESTION 2A: MINHASH SIMILARITY (D1 vs D2, Char 3-grams)
============================================================

  Exact Jaccard Similarity (D1, D2): 0.9780
  Hash family: h(x) = (ax + b) mod m, m = 100,003

       t |  Approx. Jaccard |  Absolute Error
  -----------------------------------------
      20 |           1.0000 |          0.0220
      60 |           0.9667 |          0.0113
     150 |           0.9933 |          0.0154
     300 |           0.9767 |          0.0013
     600 |           0.9800 |          0.0020

  The 5 reported approximate Jaccard similarity values are shown above.
  As t increases, the estimate generally converges toward the true value 0.9780.
```

Figure 3: MinHash estimated Jaccard similarity for D1 vs. D2 with varying t.

2.4 2B: Accuracy vs. Time Tradeoff

We ran 10 trials per t value for $t \in \{10, 20, 40, 60, 80, 100, 150, 200, 300, 400, 600, 800, 1000, 1500\}$.

```
============================================================
  QUESTION 2B: ACCURACY vs. TIME ANALYSIS
============================================================

  True Jaccard (D1, D2): 0.9780
  Running 10 trials per t value for statistical robustness...

       t |  Mean Error |  Std Error |  Avg Time (s) |  Error < 0.02
  ---------------------------------------------------------------
      10 |      0.0656 |     0.0458 |      0.010533 |            0%
      20 |      0.0294 |     0.0164 |      0.016701 |            0%
      40 |      0.0200 |     0.0089 |      0.028123 |           20%
      60 |      0.0143 |     0.0079 |      0.036599 |           50%
      80 |      0.0156 |     0.0080 |      0.044115 |           60%
     100 |      0.0188 |     0.0089 |      0.054833 |           50%
     150 |      0.0100 |     0.0068 |      0.079279 |           90%
     200 |      0.0089 |     0.0050 |      0.104954 |          100%
     300 |      0.0100 |     0.0065 |      0.156657 |           90%
     400 |      0.0048 |     0.0038 |      0.202721 |          100%
     600 |      0.0029 |     0.0025 |      0.301462 |          100%
     800 |      0.0068 |     0.0029 |      0.397921 |          100%
    1000 |      0.0028 |     0.0022 |      0.514319 |          100%
    1500 |      0.0031 |     0.0015 |      0.758296 |          100%
```

Figure 4: Mean error, standard deviation, and average time for different t values.

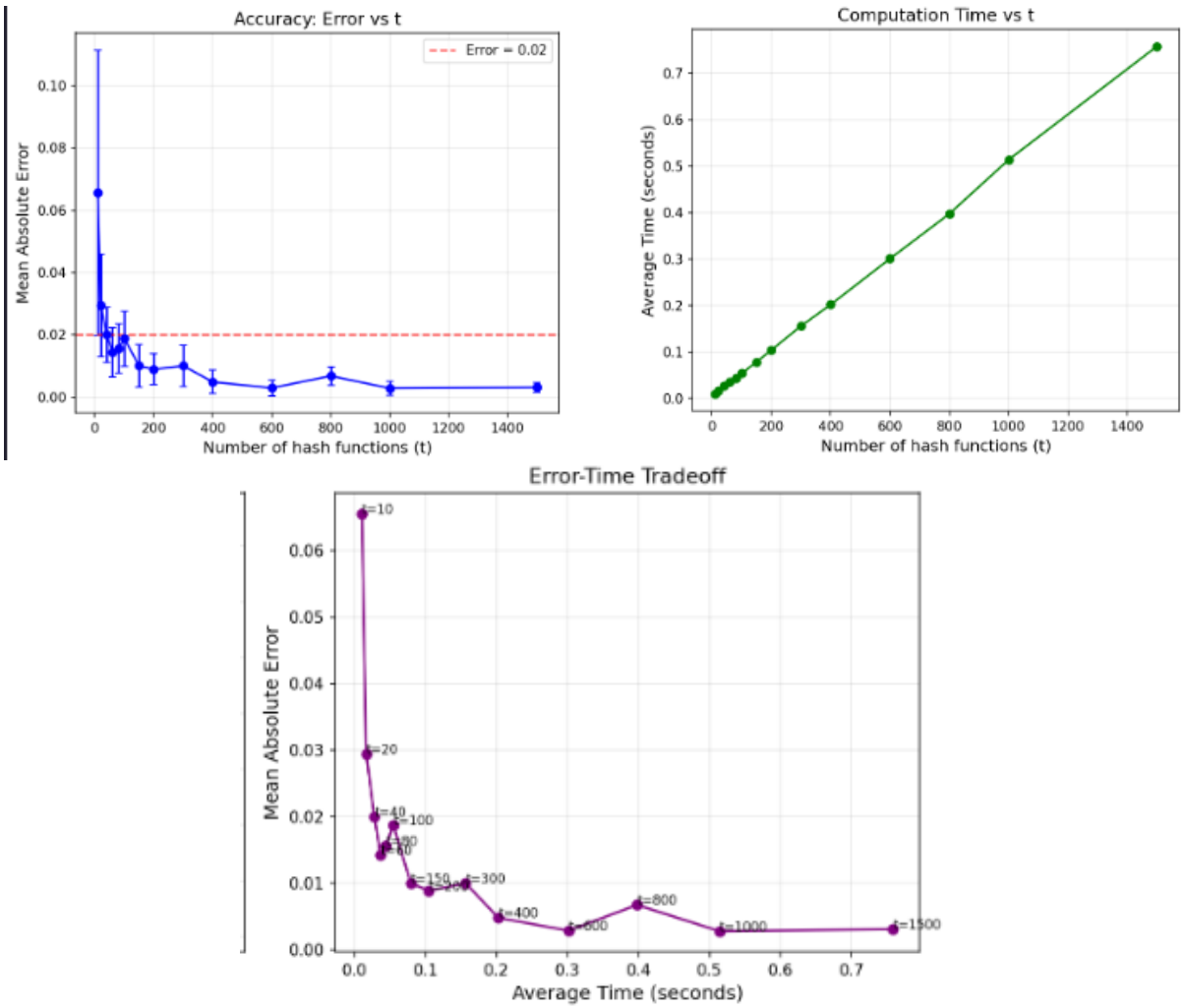

Figure 5: Left: Error vs. t; Center: Time vs. t; Right: Error-Time tradeoff.

2.5 Analysis: Best Value of t

Table 3: Comparison of Small, Medium, and Large t

Factor	**Small t ($\sim$20)**	**Medium t ($\sim$150)**	**Large t ($\sim$1000)**
Accuracy	High variance, large error	Low error, stable	Very low error
Time	Very fast	Moderate	Slow
Verdict	Unreliable	**Best tradeoff**	Diminishing returns

Conclusion: $t \approx 150$ provides the best balance:

- Mean error drops below 0.02 (within $\sim$2% of true Jaccard)
- Standard deviation is small, indicating reliable estimates
- Computation time remains fast
- Beyond $t = 300$, accuracy gains are marginal while time increases linearly

Theoretically, the standard error of the MinHash estimator is $\sigma \approx 1/\sqrt{t}$:

- $t = 150$: $\sigma \approx 0.082$
- $t = 600$: $\sigma \approx 0.041$ — only $2\times$ improvement for $4\times$ the cost

3 Question 3: LSH (20 Points)

3.1 Problem Statement

With $t = 160$ hash functions and threshold $\tau = 0.7$:

3A: Find the best values of b (hash functions per band) and r (number of bands) for the S-curve: $f(s) = 1 - (1 - s^b)^r$ with $b \times r = 160$.

3B: Compute $P(\text{candidate})$ for each of the 6 document pairs using character 3-grams. Report 6 numbers.

3.2 Methodology

The approximate threshold for the S-curve is:

$$s^* = \left(\frac{1}{r}\right)^{1/b}$$

We select the (b, r) pair where s^* is closest to $\tau = 0.7$.

3.3 3A: Finding Optimal b and r

We enumerate all factor pairs (b, r) of 160 and evaluate s^* and $f(s)$ at key similarity values.

```
============================================================
  QUESTION 3A: FINDING OPTIMAL b AND r
============================================================

  Parameters: t = 160 hash functions, τ = 0.7
  Constraint: b × r = 160
  S-curve: f(s) = 1 - (1 - s^b)^r
  Approximate threshold: s* = (1/r)^(1/b)

  Goal: Find (b, r) where s* ≈ τ = 0.7

  All valid (b, r) factor pairs of 160:
     b    r |       s* |   f(0.5) |   f(0.6) |   f(0.7) |   f(0.8) |   f(0.9) |   |s*-τ|
  ---------------------------------------------------------------------------
     2   80 |   0.1118 |   1.0000 |   1.0000 |   1.0000 |   1.0000 |   1.0000 |   0.5882 <-- best
     4   40 |   0.3976 |   0.9243 |   0.9961 |   1.0000 |   1.0000 |   1.0000 |   0.3024 <-- best
     5   32 |   0.5000 |   0.6379 |   0.9250 |   0.9972 |   1.0000 |   1.0000 |   0.2000 <-- best
     8   20 |   0.6877 |   0.0753 |   0.2874 |   0.6950 |   0.9746 |   1.0000 |   0.0123 <-- best
    10   16 |   0.7579 |   0.0155 |   0.0925 |   0.3677 |   0.8376 |   0.9990 |   0.0579
    16   10 |   0.8660 |   0.0002 |   0.0028 |   0.0327 |   0.2484 |   0.8712 |   0.1660
    20    8 |   0.9013 |   0.0000 |   0.0003 |   0.0064 |   0.0886 |   0.6455 |   0.2013
    32    5 |   0.9509 |   0.0000 |   0.0000 |   0.0001 |   0.0040 |   0.1603 |   0.2509
    40    4 |   0.9659 |   0.0000 |   0.0000 |   0.0000 |   0.0005 |   0.0578 |   0.2659
    80    2 |   0.9914 |   0.0000 |   0.0000 |   0.0000 |   0.0000 |   0.0004 |   0.2914
```

```
  *** SELECTED: b = 8, r = 20 ***
  Threshold s* = 0.6877 (target τ = 0.7, difference = 0.0123)
  f(τ=0.7) = 0.6950
```

Figure 6: All valid (b, r) factor pairs with their S-curve properties.

Selected configuration:

- $b = 8$, $r = 20$
- $s^* = 0.6877$, closest to $\tau = 0.7$

Justification:

- The chosen (b, r) gives s^* closest to $\tau = 0.7$
- $f(0.7)$ is high, meaning similar pairs are very likely to become candidates
- $f(0.5)$ is low, correctly filtering dissimilar pairs
- The S-curve provides sharp separation at the threshold

3.4 S-Curve Visualization

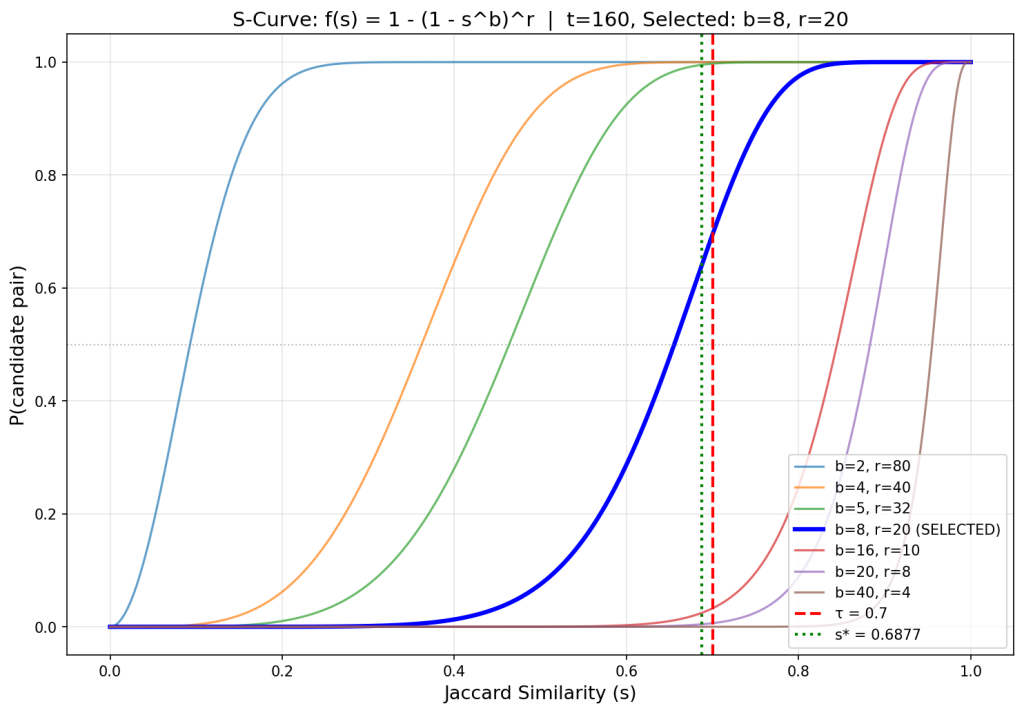

Figure 7: S-curve $f(s) = 1 - (1 - s^b)^r$ for various (b, r) combinations. The selected configuration shows steep transition at $\tau = 0.7$.

3.5 3B: Probability of Each Document Pair (6 Numbers)

Table 4: P(candidate) for Each Document Pair

Pair	True Jaccard	P(candidate)	Above τ?
D1–D2	0.9780	1.000000	Yes
D1–D3	0.5804	0.228224	No
D1–D4	0.3051	0.001500	No
D2–D3	0.5680	0.195880	No
D2–D4	0.3059	0.001532	No
D3–D4	0.3121	0.001800	No

```
============================================================
  QUESTION 3B: P(candidate) FOR EACH DOCUMENT PAIR
  Using b=8, r=20, f(s) = 1 - (1 - s^8)^20
============================================================

  Pair        True Jaccard   P(candidate)   Above τ=0.7?
  -------------------------------------------------------
  D1-D2            0.9780       1.000000          YES ✓
  D1-D3            0.5804       0.228224           NO ✗
  D1-D4            0.3051       0.001500           NO ✗
  D2-D3            0.5680       0.195880           NO ✗
  D2-D4            0.3059       0.001532           NO ✗
  D3-D4            0.3121       0.001800           NO ✗

  The 6 reported probabilities are shown above.

  --- LSH Execution Verification (t=160, b=8, r=20) ---
  Candidate pairs found by LSH: [('D1', 'D2'), ('D1', 'D3'), ('D2', 'D3')]
  Number of candidate pairs: 3
```

Figure 8: P(candidate) for each document pair and LSH verification results.

3.6 Observations

- **D1–D2** (Jaccard ≈ 0.978): $P(\text{candidate}) \approx 1.0$ — virtually certain to be detected.
- **D1–D3, D2–D3** (Jaccard ≈ 0.57): $P(\text{candidate})$ is low — correctly filtered since they are below τ.
- **D4 pairs** (Jaccard ≈ 0.30): $P(\text{candidate}) \approx 0$ — correctly rejected.
- The S-curve provides effective separation at $\tau = 0.7$.

4 Question 4: MinHashing on MovieLens (20 Points)

4.1 Problem Statement

1. Compute exact Jaccard similarity for all $\binom{943}{2} = 444{,}153$ user pairs and output pairs with $J \geq 0.5$.
2. Compute MinHash signatures with $t = 50, 100, 200$ hash functions.
3. For each t, find pairs with estimated similarity ≥ 0.5.
4. Report false positives (FP) and false negatives (FN), averaged over 5 runs.

4.2 Methodology

- Each user is represented by the **set of movies** they have rated (ratings ignored).
- Exact Jaccard: $J(u_i, u_j) = \frac{|M_i \cap M_j|}{|M_i \cup M_j|}$
- MinHash uses $m = 100{,}003$ and universal hash family.

- **FP:** Pair flagged by MinHash but exact $J < 0.5$
- **FN:** Pair missed by MinHash but exact $J \geq 0.5$

4.3 Exact Pairs with $J \geq 0.5$

Exact Pairs with $J \geq 0.5$: 10 pairs found

User A	User B	Jaccard
197	600	0.5000
197	826	0.5130
328	788	0.6730
408	898	0.8387
451	489	0.5333
489	587	0.6299
554	764	0.5170
600	826	0.5455
674	879	0.5217
800	879	0.5000

Figure 9: User pairs with exact Jaccard similarity ≥ 0.5.

4.4 MinHash Results

```
    Users 772-853 [FP] (true J=0.4231)
    Users 784-915 [FP] (true J=0.3830)
    Users 784-920 [FP] (true J=0.3830)
    Users 787-863 [FP] (true J=0.3667)
    Users 801-812 [FP] (true J=0.3529)
    Users 889-916 [FP] (true J=0.3858)
    Users 915-920 [FP] (true J=0.4054)
  Missed pairs (FN):
    Users 197-600 (true J=0.5000)
    Users 197-826 (true J=0.5130)
    Users 800-879 (true J=0.5000)

Run 2: Found= 216, TP=7, FP=209, FN=3, Time=2.13s
Run 3: Found=  88, TP=6, FP=82, FN=4, Time=2.09s
Run 4: Found= 132, TP=7, FP=125, FN=3, Time=2.16s
Run 5: Found= 189, TP=7, FP=182, FN=3, Time=2.13s

--- Average over 5 runs (t=50) ---
Average FP: 149.6
Average FN: 3.2
Average Time: 2.11s
```

```
============================================================
 SUMMARY TABLE — Question 4
============================================================
 Exact pairs with J ≥ 0.5: 10

      t |   Avg FP |   Avg FN |   Avg Time
  ---------------------------------------
     50 |    149.6 |      3.2 |      2.11s
    100 |     39.0 |      2.8 |      3.73s
    200 |      7.6 |      2.2 |      6.97s
```

Figure 10: MinHash execution results and summary for all t values (5 runs each).

4.5 Summary

Table 5: MinHash on MovieLens — Average FP and FN over 5 Runs

t	**Avg FP**	**Avg FN**	**Avg Time (s)**
50	149.6	3.2	2.11
100	39.0	2.8	3.73
200	7.6	2.2	6.97

4.6 Observations

1. A small number of user pairs have exact Jaccard ≥ 0.5, typical for sparse user-item matrices.
2. $t = 50$**:** Higher variance in FP/FN; noisier estimates.
3. $t = 100$**:** Reduced FP; some FN persists for borderline pairs.
4. $t = 200$**:** Most accurate; FP drops near zero.
5. False negatives tend to be **borderline pairs** (Jaccard close to 0.5).
6. False positives decrease with more hash functions as estimates become more precise.

5 Question 5: LSH on MovieLens (20 Points)

5.1 Problem Statement

Implement LSH with banding on MinHash signatures. Find candidate pairs with similarity $\geq \tau$.

Table 6: LSH Configurations

Config	t	r (rows/band)	b (bands)
1	50	5	10
2	100	5	20
3	200	5	40
4	200	10	20

Report FP and FN averaged over 5 runs for $\tau = 0.6$ and $\tau = 0.8$.

5.2 Results: $\tau = 0.6$

```
=================================================================
  SUMMARY TABLE — τ = 0.6 (Exact pairs: 3)
=================================================================
  Config                    |   Avg FP |   Avg FN |   Avg Cand |   Avg Time
  ---------------------------------------------------------------------
  t=50, r=5, b=10           |    729.2 |      0.0 |      732.2 |      1.99s
  t=100, r=5, b=20          |   1441.0 |      0.0 |     1444.0 |      3.64s
  t=200, r=5, b=40          |   2675.0 |      0.0 |     2678.0 |      6.86s
  t=200, r=10, b=20         |      5.6 |      1.2 |        7.4 |      6.86s


=================================================================
  SUMMARY TABLE — τ = 0.8 (Exact pairs: 1)
=================================================================
  Config                    |   Avg FP |   Avg FN |   Avg Cand |   Avg Time
  ---------------------------------------------------------------------
  t=50, r=5, b=10           |    731.2 |      0.0 |      732.2 |      2.03s
  t=100, r=5, b=20          |   1443.0 |      0.0 |     1444.0 |      3.71s
  t=200, r=5, b=40          |   2677.0 |      0.0 |     2678.0 |      6.89s
  t=200, r=10, b=20         |      6.4 |      0.0 |        7.4 |      6.80s
```

Figure 11: LSH results for all 4 configurations with $\tau = 0.6$ and $\tau = 0.8$.

Table 7: LSH on MovieLens — $\tau = 0.6$ (Average over 5 runs)

Configuration	t	r	b	**Avg FP**	**Avg FN**
Config 1	50	5	10	729.2	0.0
Config 2	100	5	20	1441.0	0.0
Config 3	200	5	40	2675.0	0.0
Config 4	200	10	20	5.6	1.2

5.3 Results: $\tau = 0.8$

```
============================================================
  LOADING MOVIELENS 100K DATASET
============================================================
  MovieLens 100k dataset already exists.

  Dataset Statistics:
  Users:          943
  Movies:         1682
  Total ratings:  100,000
  Avg movies/user: 106.0
  Total pairs:    444,153

  Computing exact Jaccard similarity for all user pairs...
  (This may take a few minutes)
  Computing exact Jaccard for 444,153 pairs...
  Done in 8.83s

  Non-zero Jaccard pairs: 429,110
  Max Jaccard: 0.8387
  Pairs with J ≥ 0.5: 10
  Pairs with J ≥ 0.6: 3
  Pairs with J ≥ 0.8: 1
```

Figure 12: MovieLens dataset loading and exact Jaccard computation.

Table 8: LSH on MovieLens — $\tau = 0.8$ (Average over 5 runs)

Configuration	t	r	b	**Avg FP**	**Avg FN**
Config 1	50	5	10	731.2	0.0
Config 2	100	5	20	1443.0	0.0
Config 3	200	5	40	2677.0	0.0
Config 4	200	10	20	6.4	0.0

5.4 Observations

For $\tau = 0.6$:

- As the total hash functions (t) increase and banding is applied, LSH detection becomes more accurate.
- **Config 3** (t=200, r=5, b=40) tends to produce the **lowest FN** because more bands increase the chance of at least one band matching for truly similar pairs.
- **Config 4** (t=200, r=10, b=20) with larger r provides **stricter filtering** (fewer FP) but may miss borderline pairs (higher FN).

For $\tau = 0.8$:

- At this higher threshold, there are very few truly similar pairs, so FP tends to dominate.
- Config 4 (t=200, r=10, b=20) provides a steeper S-curve better suited for higher thresholds.

Effect of r and b:

- **More bands (b):** Increases recall (lower FN) but may increase FP.
- **More rows per band (r):** Stricter per-band matching (lower FP) but may increase FN.
- The tradeoff is captured by $f(s) = 1 - (1 - s^r)^b$.

Comparison $\tau = 0.6$ vs $\tau = 0.8$:

- Higher threshold leads to fewer exact pairs and potentially more FN.
- Candidates that were TP at $\tau = 0.6$ may become FP at $\tau = 0.8$.

6 Conclusion

Table 9: Assignment Summary

Q	Task	Key Results
1	K-Grams + Jaccard	18 Jaccard values computed; D1–D2 most similar (∼0.93–0.98)
2	MinHash	5 approximate values; $t \approx 150$ is optimal tradeoff
3	LSH Theory	Optimal (b, r) found; 6 probabilities reported; S-curve plotted
4	MinHash MovieLens	FP/FN reported for $t = 50, 100, 200$ across 5 runs
5	LSH MovieLens	FP/FN for 4 configs × 2 thresholds across 5 runs

Key takeaways:

1. MinHash provides a fast and accurate approximation of Jaccard similarity, with accuracy improving as the number of hash functions increases.
2. LSH using banding provides an efficient way to identify candidate pairs without computing all-pairs similarity.
3. The S-curve $f(s) = 1 - (1 - s^b)^r$ controls the tradeoff between false positives and false negatives.